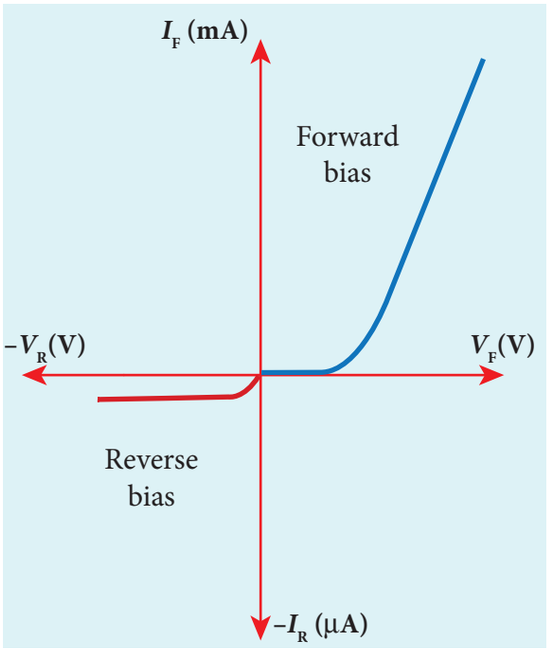
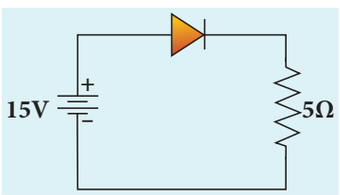



**Figure 10.16** Forward and reverse charactristics of a diode

## EXAMPLE 10.1




An ideal diode and a 5 Ω resistor are connected in series with a 15 V power supply as shown in figure below. Calculate the current that flows through the diode.




### Solution

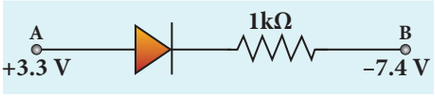
The diode is forward biased and it is an ideal one. Hence, it acts like a closed switch with no barrier voltage. Therefore, current that flows through the diode can be calculated using Ohm's law.

$$V = IR$$

$$I = \frac{V}{R} = \frac{15}{5} = 3\,\text{A}$$

## EXAMPLE 10.2

A silicon diode is connected with 1kΩ resistor as shown. Find the value of current flowing through *AB*.




### Solution

The P.D. between A and B is given by

$$V = [V_A - V_B] - V_b(\text{Si})$$
$$= [3.3 - (-7.4)] - 0.7$$
$$= 10.7 - 0.7 = 10\ \text{V}$$

The value of current flowing through *AB* can be obtained using Ohm's law.

$$I = \frac{V}{R} = \frac{10}{1\times10^3} = 10^{-2}\,\text{A} = 10\,\text{mA}$$

## 10.3.4 Rectification

**The process in which alternating voltage or alternating current is converted into direct voltage or direct current is known as rectification**. The device used for this process is called as rectifier. In this section, we will discuss two types of rectifiers namely, half wave rectifier and full wave rectifier

### i) Half wave rectifier circuit

The half wave rectifier circuit consists of a transformer, a *p-n* junction diode and a resistor (Figure 10.17(a)). In a half wave rectifier circuit, either a positive half or the negative half of the AC input is passed through by the diode while the other half is blocked. Only one half of the input wave is rectified. Therefore, it is called half wave rectifier. Here, a *p-n* junction diode acts as a rectifier diode.

*During the positive half cycle*

When the positive half cycle of the AC input signal passes through the circuit,



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